

MD.MOINUL ISLAM

Curriculum Vitae

Block E, Road: 9, Section: 1, Mirpur
Dhaka – 1216, Bangladesh
☎ (+88) 01812089547
✉ islam2205101592@diu.edu.bd
📄 moinul909.github.io
🐙 Github in LinkedIn



About me

Computer Science and Engineering graduate from Daffodil International University with research interests in Artificial Intelligence, Digital Health, and Trustworthy AI. My expertise includes Machine Learning, Natural Language Processing, Computer Vision, Human–Computer Interaction, Data Mining, Data Management, Health Information Systems, Information Systems, Computer Security, Data Privacy and AI Governance. As a Research Associate & Coordinator at the Health Informatics Research Lab (HIRL), I contribute to AI-driven healthcare research, project coordination, and researcher mentorship. I also worked as a Generative AI Intern at Metademy LLC, USA, developing Large Language Model (LLM)-based applications and AI agents. My research focuses on building intelligent, privacy-preserving, and clinically interpretable AI systems for healthcare and digital health applications.

Education

- 2022–2026 **B.Sc. in Computer Science & Engineering**, *Daffodil International University*, Dhaka.
CGPA: 3.91/4.00 Graduation Date: 27 July 2026
Thesis: “Beat by Beat Lead Aware ECG Segmentation and Multi Class Cardiac Disease Classification Using the PTB Diagnostic Database” (Sep 2025 – Aug 2026, 6 credits)
Supervisor: Dr. Md Zahid Hasan, Associate Professor & Director, HIRL
- 2021 **Higher Secondary Certificate (HSC), Science**, *Kushtia Govt. Central College*, Kushtia.
GPA: 5.00/5.00
- 2019 **Secondary School Certificate (SSC), Science**, *Nabagram Secondary School*, Kushtia.
GPA: 5.00/5.00

Publications

Under Review / Submitted

- 2025 **Md. Moinul Islam**, Sadia Sultana Chowdhury, Md. Zahid Hasan, and Sami Azam, “Beat-by-Beat, Lead-aware ECG Segmentation and Subject-independent Myocardial Infarction Detection.”
- 2026 **Md. Moinul Islam**, Ashabari Mayurakkhi, Rokeya Unzum Shekha, Md. Zahid Hasan, Md Jamil Istiaq, and Dewan Aminul Islam, “IKDEFL: An Independent Knowledge Distillation Enhanced Federated Learning Framework for Ear Disease Detection.” Submitted to: COMPAS 2026, Submission ID: 386, under review (2026), regional.

Research in Progress

- 2026–present CoughScreen-BD: A Non-Invasive Acoustic Deep Learning Framework for cough-based disease screening. (*Data collection / model development stage; manuscript not yet drafted.*)
- 2026–present Clinically Explainable Visual Question Answering (X-VQA) for Gastrointestinal Endoscopy. (*Model development stage; manuscript not yet drafted.*)

Research & Professional Experience

Daffodil International University

- Feb 2026 – **Research Associate & Coordinator**, *Health Informatics Research Lab (HIRL)*.
present Employment status: Part-time, 35/week; Supervisor: Dr. Md. Zahid Hasan, HIRL
Coordinating research projects, managing research teams, mentoring students, and supporting academic collaborations and research programs.
- Fall 2025 – **Undergraduate Teaching Assistant**, *Dept. of CSE, Daffodil International University*.
- Summer 2026 Employment status: Part-time, 30/week, Supervisor: Dr. S. M. Aminul Haque
Assisted in preparing and delivering course materials for undergraduate CSE courses and student consultations.
- Jan 2025 – **Research Assistant (RA)**, *Health Informatics Research Lab (HIRL)*.
- Feb 2026 Employment status: Part-time, 20/week; Supervisor: Dr. Md. Zahid Hasan, HIRL
Contributed to data preprocessing, model development, and experimental evaluation for AI-in-healthcare projects.

Industry Experience

- Apr 2025 – **Generative AI Intern**, *Metademy LLC, Utah, United States*.
- Jul 2025 Employment status: Part-time, 30/week, Supervisor: Md. Uday Ahmed
Contributed to AI research and product innovation using LLMs, Whisper API, and AI agents.

Teaching Experience

- Spring 2026 – **Instructor, PythonX: Python Foundations for Data Science Boot Camp**, *HIRL Lab*.
- Summer 2026 Conducted hands-on Python and Data Science training sessions.
- Jul 2024 – **Instructor, Data Structures and Algorithms (DSA)**, *Q-bit Learning*.
- Dec 2024 Conducted classes on Data Structures, Algorithms, and problem-solving techniques for students.

Leadership & Service

- Apr 2026 **Organizer**, *Research Policy Initiative*, Department of Computer Science & Engineering, Daffodil International University.
- Spring 2026 **Course Module Leader**, *PythonX: Python Foundations for Data Science Boot Camp*, Health Informatics Research Lab (HIRL).
- Feb 16–22, 2026 **Co-Organizer**, *Research Workshop: From Zero to First Idea*, Division of Research & Health Informatics Research Lab (HIRL).
- Apr 2025 – **President**, *Health Informatics Research Lab (HIRL)*.
Apr 2026
- Mar 2025 – **Vice President, Research & Journal Wing**, *Daffodil International University Computer Programming Club (DIU CPC)*.
Apr 2026
- Mar 2024 – **Joint Secretary**, *Health Informatics Research Lab (HIRL)*.
Mar 2026

Honours & Awards

- Oct 2025 Judge and Problem Setter, **Prompt Battle Fall 25**, Daffodil International University CPC Club.
- Aug 2025 **Dean's List Award**, Fall 2024, Daffodil International University.
- May 2025 Judge and Problem Setter, **Prompt Battle Summer 25**, Daffodil International University CPC Club.
- Oct 2024 **Library Excellence Award**, Daffodil International University Library.
- Jun 2024 **Dean's List Award**, Fall 2023, Daffodil International University.
- 2019 **Certificate of Appreciation** from Zilla Parishad, Kushtia for outstanding performance in the JSC/JDC/SSC Equivalent Examination 2019.

- 2016 **1st Merit Position** in the Annual Examination 2016 at Rahmatpur Government Primary School, Daulatpur, Kushtia.
- 2013 **Talentpool Scholarship**, Primary Education Completion Examination (PECE), Daulatpur Upazila, Kushtia, Bangladesh.
- 2013 **8th Position (Upazila Merit List)**, Primary Education Completion Examination (PECE), Kushtia, 2013.

Certificates & Training

- Dec 2025 Volunteer, *International Conference on Intelligent Data Analysis and Applications (IDAA 2025)*, December 12–13, 2025.
- Nov 2025 Participant, *FutureNation Summit 2025*, National Science and Technology Complex, Dhaka, November 30, 2025.
- Nov 2025 *Employability Masterclass 21*, UNDP Bangladesh, November 11, 2025.
- Nov 2025 *English Online Self-Study: Social Interaction (Level B1)*, British Council, November 10, 2025.
- Sep 2025 *Preparatory Online Training on IELTS Speaking Module*, FutureNation UNDP Bangladesh, September 03, 2025.
- Jul 2025 *Industry-Academia Collaboration Summit 2025*, Career Development Center (CDC), Daffodil International University, July 12–13, 2025.
- May 2025 Participate *DIU-DOR Poster Presentation 2025*, May 14, 2025.
- Apr 2025 *Linux Server Management and Security*, University of Colorado System via Coursera, April 02–27, 2025.
- Apr 2025 *Data Science Math Skills*, Duke University via Coursera, April 10–23, 2025.
- Mar 2025 *Automation in the AWS Cloud*, Amazon Web Services (AWS) via Coursera, March 27 – April 01, 2025.

Research Skills

- Machine Learning & Deep Learning Practical experience designing, training, and evaluating machine learning and deep learning models, including CNNs, Transformers, and Vision Transformers, for classification, segmentation, and detection tasks.
- Federated Learning Hands-on experience developing privacy-preserving federated learning frameworks with knowledge distillation for decentralized model training.
- Explainable AI Applied explainability techniques, including attention visualization and feature attribution, to develop interpretable and clinically meaningful AI models.
- NLP & LLMs Experience developing LLM-based applications and autonomous AI agents using prompt engineering, retrieval-augmented workflows, and speech-to-text integration with Whisper API.
- Computer Vision Research experience in medical image analysis, including endoscopic imagery and ear-disease image classification using deep learning and vision-based models.
- Medical Data Analysis Experience analyzing physiological and clinical data, including ECG signal segmentation, cardiac disease classification, and acoustic cough-signal analysis for disease screening.
- Multimodal Learning Experience integrating and analyzing multimodal medical data, including medical images, clinical text, and physiological signals for healthcare AI applications.
- Experimental Evaluation Experience designing rigorous experiments using subject-independent data splits, cross-validation, ablation studies, and standard classification and segmentation metrics to assess model performance and generalizability.

Research Frameworks & Tools Python, PyTorch, TensorFlow, Scikit-learn, Jupyter Notebook, Google Colab, Kaggle, AWS, Linux, Anaconda.

Data Analysis & Visualization Pandas and NumPy for data processing and statistical analysis; Matplotlib, Seaborn, and Plotly for research-oriented visualization, exploratory analysis, and presentation of experimental results.

Languages

Bangla **Mother Tongue**

English **Fluent in Reading, Listening, and Writing; Conversational in Speaking**

English **IELTS planned for 10 September 2026**

Proficiency Test

Referees

Dr. Sami Azam

Professor

Information Technology

IEEE NT Student Activities Chair

Charles Darwin University, Australia

✉ Sami.azam@edu.edu.au

🌐 Personal Webpage

Dr. SM. Aminul Haque

Professor & Associate Head

Department of Computer Science & Engineering

Faculty of Science and Information Technology

Daffodil International University, Bangladesh

📞 +88 01847140129

✉ aheadcse2@daffodilvarsity.edu.bd

🌐 Personal Webpage

Dr. Md Zahid Hasan

Associate Professor

Department of Computer Science & Engineering

Faculty of Science and Information Technology

Daffodil International University, Bangladesh

📞 +88 01672580748

✉ zahid.cse@diu.edu.bd

🌐 Personal Webpage